

Software Requirements Specification (SRS)

SmartFarm Management Application

Document Version: 1.0	SDLC Stage: Phase 1 — Requirements Specification & Compliance Expansion	Status: Approved Baseline
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document details the complete functional, non-functional, security, and architectural requirements for the **SmartFarm Management Application**. It serves as the authoritative single source of truth for engineering, compliance testing, and continuous maintenance.

1.2 Scope of the System

SmartFarm is an enterprise-grade cloud farm management platform designed for modern agricultural enterprises. The application enables:

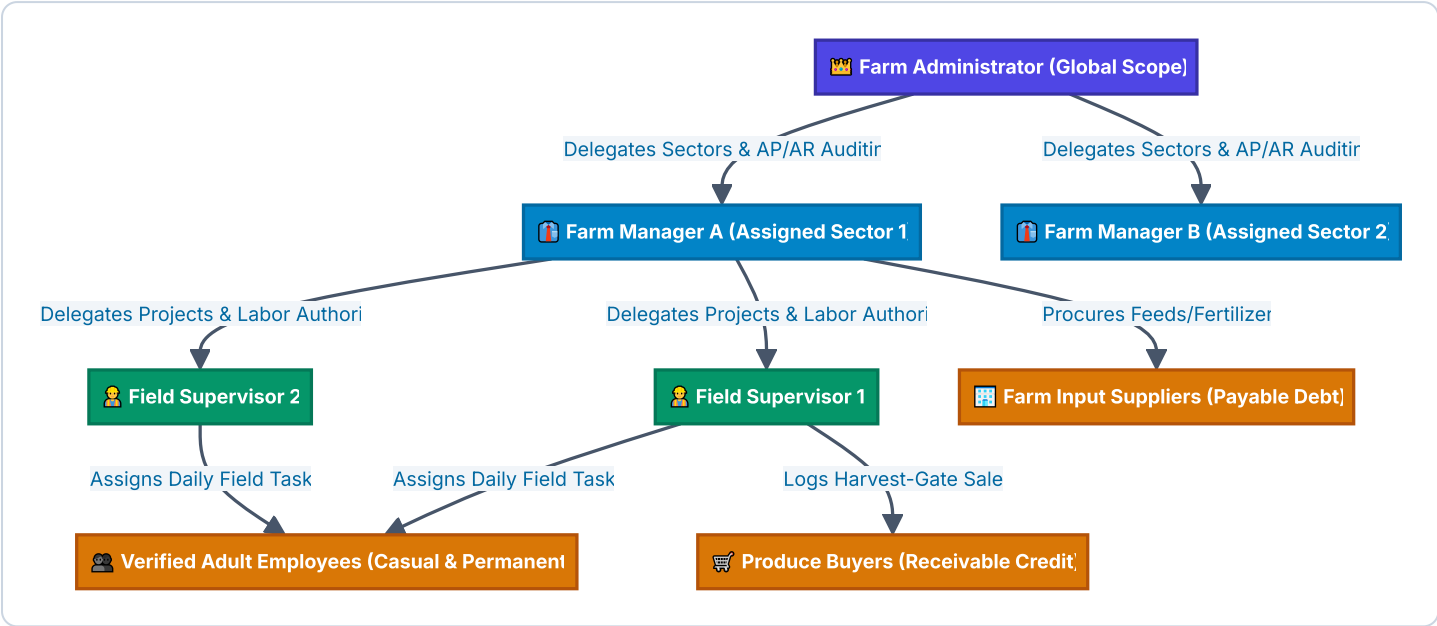
- Executive governance, labor auditing, and financial/debt oversight by **Farm Administrators**.
- Sector-level resource management, project planning, supplier procurement, and supervisor delegation by **Farm Managers**.
- Field-level task coordination, harvest logging, input consumption, and verified laborer work assignment by **Field Supervisors** under strict financial privacy isolation.
- Comprehensive tracking of external trade relationships: **Suppliers (Accounts Payable / Farm Debt)** and **Customers (Accounts Receivable / Customer Credit)**.

1.3 Definitions, Acronyms, and Abbreviations

- RBAC:** Role-Based Access Control.
- PBAC:** Privilege/Policy-Based Access Control (fine-grained feature toggle permissions).
- Labor Compliance:** Strict validation of government National ID cards to guarantee legal adult employment eligibility (\$Age \ge 18\$) and eliminate child labor risks.
- Accounts Payable (AP):** Cumulative financial liabilities owed by the farm to external input/feed/equipment suppliers.
- Accounts Receivable (AR):** Cumulative credit balances owed to the farm by harvest and produce buyers.
- Zero-State Dashboard:** A tailored, clean UI rendered when an authenticated user has zero assigned projects/categories, suppressing empty graphs and displaying actionable next steps.

2. Overall Description & System Context

2.1 System Actors & Hierarchy



Persona / Entity	System Scope	Max Quota	Core Responsibilities
Farm Administrator	Entire Farm (Global)	1 Account	Global user provisioning, AP/AR debt auditing, category allocation, system governance.
Farm Manager	Assigned Categories	Max 2 Accounts	Project budgeting, supervisor provisioning, supplier procurement, labor auditing.
Field Supervisor	Assigned Projects	Max 10 Accounts	Daily field activity execution, harvest logging, worker task allocation, stock usage.
Employees / Laborers	Assigned Tasks	Scalable Registry	Verified adult workforce (\$Age \ge 18\$, National ID required) executing field operations.
Suppliers	Farm Inputs Catalog	Scalable Directory	Vendors providing seed, fertilizer, animal feed, tools, and machinery on cash/credit terms.
Customers	Produce Distribution	Scalable Directory	Wholesale/retail buyers purchasing farm yields on cash, M-Pesa, or credit terms.

3. Detailed Functional Requirements (FR)

FR-1: Authentication, Account Provisioning & Password Recovery

FR-1.1: System shall support authentication using either `Username` or `Email` combined with a password.

FR-1.2: Unauthenticated users can request a password reset by providing their registered email address (`POST /users/forgot-password`).

FR-1.3: If an email is not registered in the system, the API shall return HTTP 400 with: `"No account found with this email address. If you did not register an email, please contact your Farm Administrator to reset your password."`

FR-1.4: If the email exists, the system generates a secure cryptographic token (`PasswordResetToken` valid for 1 hour) and sends an HTML reset email via SMTP.

FR-1.5: Administrators can execute direct password resets for staff accounts via the User Management portal (`PATCH /users/{id}/admin-reset-password`).

FR-2: Staff Quotas & Hierarchical Access Delegation (PBAC)

FR-2.1: The system shall strictly enforce staff limits: Max 1 Admin, Max 2 Managers, Max 10 Supervisors.

FR-2.2: Admin to Manager Delegation:

- Admin assigns one or more Farm Categories to a Manager.
- Admin configures fine-grained privilege toggles for the Manager:
 - `CAN_CREATE_CATEGORIES` : Create and configure new farm sectors.
 - `CAN_CREATE_SUPERVISORS` : Hire and provision supervisor accounts.
 - `CAN_VIEW_FINANCIALS` : View revenue, expense, and cash flow analytics.
 - `CAN_MANAGE_BUDGETS` : Create projects and modify budget allocations.
 - `CAN_DELETE_INVENTORY` : Remove catalog items from inventory.

FR-2.3: Manager to Supervisor Delegation:

- Manager provisions supervisors linked to their manager account ID (`manager_id`).
- Manager assigns specific projects (within the Manager's categories) to the Supervisor.
- Manager configures fine-grained privilege toggles for the Supervisor:
 - `CAN_RECORD_HARVEST` : Submit harvest volumes and production logs.
 - `CAN_LOG_ACTIVITIES` : Log daily field activities and task operations.
 - `CAN_USE_INVENTORY` : Deduct fertilizer, feed, and seed quantities from inventory.
 - `CAN_RECORD_EXPENSES` : Log petty cash expenses incurred in the field.
 - `CAN_RECORD_SALES` : Record direct farm-gate sales without global financial access.

FR-3: Workload & Capacity Validation Rules

FR-3.1 (Manager Workload): When assigning a category to a Manager, if assignments exceed 3 active categories, an advisory workload confirmation is triggered.

FR-3.2 (Supervisor Workload): Each supervisor has a `max_project_capacity` (default 4 projects). The system blocks assigning additional active projects unless capacity is adjusted or existing projects complete.

FR-4: Category & Project Lifecycle Management

FR-4.1: Projects must belong to an existing Category and possess a globally unique project name.

FR-4.2 (Date Validation Rule): The system shall strictly validate that `startDate ≤ endDate`. If `startDate > endDate`, both backend and frontend reject the request with: `"Start date cannot be greater than end date!"`

FR-4.3 (Budget Modification Rule): Modifying project details, season, dates, or budgets strictly requires `CAN_MANAGE_BUDGETS` privilege.

FR-5: Operational Field Tracking (Harvest, Activity, Inventory)

FR-5.1: Harvest records capture item name, numerical yield, unit of measurement (KG, Litres, Bags), date, and notes.

FR-5.2: Activities capture activity title, type (Weeding, Irrigation, Spraying, Feeding, Pruning), date, notes, and attached laborers.

FR-5.3: Inventory tracks quantity in stock, unit prices, reorder levels, and records deduction history when consumed by projects.

FR-6: Financial Governance & Sales Enforcement

FR-6.1 (Stock-Backed Sales Rule): A project cannot record a sale exceeding its total recorded harvest yields minus previous sales.

FR-6.2 (Supervisor Financial Shield): Supervisor API endpoints and dashboard views strip budget allocations, sales revenue, and profit calculations.

FR-7: Dynamic Scoped Dashboards & Zero-State Engine

FR-7.1 (Admin Dashboard): Renders enterprise-wide aggregates, category financial distribution, user quotas, and global cashflow.

FR-7.2 (Manager Dashboard): Aggregates budget, project counts, and supervisor workloads strictly for the manager's assigned categories.

FR-7.3 (Supervisor Dashboard): Aggregates harvest volumes, pending tasks, and recent logs strictly for assigned projects.

FR-7.4 (Zero-State Suppression): If a user has 0 assigned categories or 0 assigned projects:

- All metrics cards, financial figures, and graphs are **suppressed**.
- Renders a dedicated **Action Hero Card** providing immediate contextual instructions (e.g., *"No categories assigned yet. Contact your Farm Administrator to get started."*).

FR-8: User Deletion & Referential Integrity Safeguards

FR-8.1: Admin accounts cannot be deleted (`isUserAdmin` protection).

FR-8.2: Deleting a user safely cascades/unlinks child records: deletes active `PasswordResetToken` rows, clears category join tables, unlinks `supervisor_id` on projects, and clears `createdById` references.

FR-9: Labor Compliance, Legal Adult Verification & Work Assignment



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FR-9.1 (Adult Labor Verification): All farm hands and field workers must be registered with their official government **National ID number**, **Full Name**, and **Phone Number** (for identification and M-Pesa payroll).

FR-9.2 (Minor Employment Prohibition): System strictly blocks onboarding laborers without verified government identity credentials, preventing child labor compliance violations.

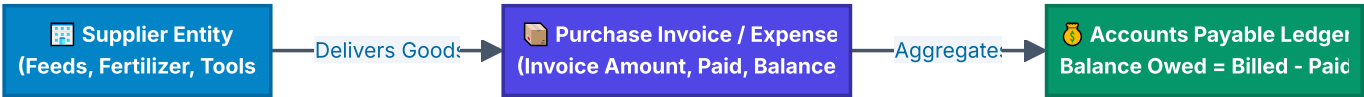
FR-9.3 (Employment Classification): Supports `CASUAL` (daily wage / task-based) and `PERMANENT` (salaried / monthly) employee profiles.

FR-9.4 (Task & Activity Allocation):

- Admins, Managers, and Field Supervisors can assign one or more verified employees to any project activity (e.g., **Weeding Sector 1**, **Spraying Block C**, **Irrigation Line Setup**).
- Captures: `activity_id`, `employee_id`, `date`, `hours_worked`, `daily_rate`, `wage_payable`, and supervisor remarks.

FR-9.5 (Labor Audit Trail): Provides a complete historical log of all tasks performed by an employee across farm projects for labor auditing and wage settlements.

FR-10: Supplier Management & Accounts Payable (Debt) Ledger



FR-10.1 (Supplier Registry): Centralized directory of external farm suppliers capturing: `name`, `contact_person`, `phone_number`, `email`, `id_or_tax_number` (KRA PIN / National ID), `category_supplied` (**Fertilizer & Agro-chemicals**, **Animal Feeds**, **Seeds & Seedlings**, **Farm Machinery & Hardware**, **Veterinary Supplies**), and physical address.

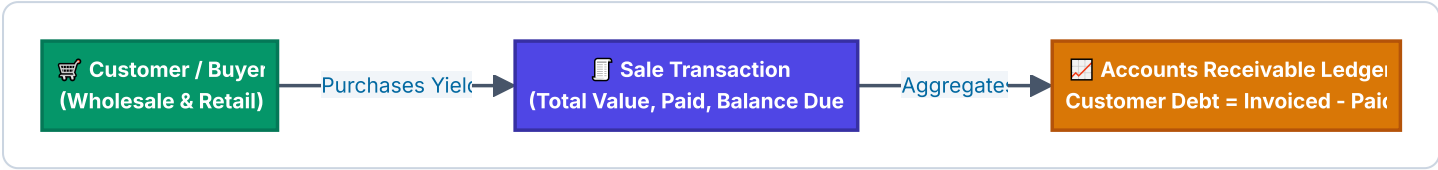
FR-10.2 (Purchase & Expense Linkage): When stock is added to inventory or project cash expenses are recorded, transactions can be linked directly to a registered `supplier_id`.

FR-10.3 (Accounts Payable / Debt Tracking):

- For each supplier transaction, captures: `invoice_number`, `invoice_amount`, `amount_paid`, `balance_due`, `payment_status` (`PAID`, `PARTIALLY_PAID`, `CREDIT_UNPAID`), and `payment_due_date`.
- Calculates live supplier balances: $\text{\$}\{Balance\ Owed\} = \sum \{Invoice\ Amount\} - \sum \{Amount\ Paid\}\$$.

FR-10.4 (Executive Debt Governance): Admin and authorized Managers have dedicated dashboard widgets tracking Total Farm Accounts Payable (Outstanding Debt Owed to Suppliers) with overdue liability alerts.

FR-11: Customer Credit Governance & Accounts Receivable Ledger



FR-11.1 (Customer Registry Expansion): Stores buyer identity including `name` , `contact` , `id_number` , `address` , `credit_limit` , and `credit_status` (`CLEAR` , `HAS_DEBT` , `CREDIT_BLOCKED`).

FR-11.2 (Credit Sales & Partial Payments):

- When recording sales, supports immediate full cash/M-Pesa payment, partial payment, or full credit terms.
- Captures: `total_amount` , `amount_paid` , `balance_due` ($\text{\$total_amount} - \text{\$amount_paid}$), `payment_mode` (`CASH` , `MPESA` , `BANK_TRANSFER` , `CREDIT_LEDGER`), and `payment_status` (`PAID_IN_FULL` , `PARTIAL_PAYMENT` , `CREDIT_UNPAID`).

FR-11.3 (Accounts Receivable / Debtor Tracking):

- Automatically calculates customer debt balances: $\text{\$Outstanding Debt} = \sum \text{\$Total Invoiced} - \sum \text{\$Total Remitted}$.
- Flags customers exceeding credit terms or with overdue balances during new sale attempts.

FR-11.4 (Supervisor Privacy Isolation): Field Supervisors can log transaction-level cash/credit splits for direct farm-gate sales without gaining access to cumulative farm-wide debtor reports.

4. Non-Functional Requirements (NFR)

- NFR-1 (Security):** BCrypt password hashing ($\text{\$cost} = 10$), role/privilege-based endpoint security, parameter sanitization, and SQL injection prevention via Spring Data JPA.
- NFR-2 (Performance):** Page load times $\text{\$} < 1.5\text{s}$ on broadband; database indexed on `username` , `email` , `category_id` , `supervisor_id` , `manager_id` , `employee_id` , `supplier_id` , `customer_id` .
- NFR-3 (Reliability):** 99.9% uptime target backed by automated heartbeat monitoring.
- NFR-4 (Cost):** 100% compliant with zero-cost cloud architecture limits (Render Free Web Service, Vercel Hobby, TiDB Cloud / Aiven Free Tier).
- NFR-5 (Usability):** Modern design system with responsive layouts, accessible color contrast, and dedicated mobile field data entry views.